(iii) $(-2)^5 \times (-2)^{-3} = (-2)^{5-3} = (-2)^2 = -2 \times -2 = 4$ (iv) $\frac{3^2}{3^{-2}} = 3^2 \times 3^2 = 9 \times 9 = 81$

Example 1.37

Simplify and write the answer in exponential form:

(i) $(3^5 \div 3^8)^5 \times 3^{-5}$ (ii) $(-3)^4 \times \left(\frac{5}{3}\right)^4$

Solution:

(i) $\left(\frac{3^5}{3^8}\right)^5 \times 3^{-5} = (3^{5-8})^5 \times 3^{-5} = (3^{-3})^5 \times 3^{-5} = 3^{-3\times5} \times 3^{-5} = 3^{-15} \times 3^{-5} = 3^{-15-5} = 3^{-20}$

(ii) $(-3)^4 \times \left(\frac{5}{3}\right)^4 = 3^4 \times \frac{5^4}{3^4} = 5^4$

Example 1.38

Find x so that $(-7)^{x+2} \times (-7)^5 = (-7)^{10}$

Solution:

$$(-7)^{x+2} \times (-7)^5 = (-7)^{10}$$

$$(-7)^{x+2+5} = (-7)^{10}$$

Since the bases are equal, we equate the exponents to get

$$x + 7 = 10$$

$$\Rightarrow x = 10 - 7 = 3$$

1.9.5 Standard Form and Scientific Notation

Standard form of a number is just the number as we normally write it. We use expanded notation to show the value of each digit. That is, it is exhibited as a sum of each digit duly multiplied by its matching place value (like ones, tens, hundreds etc.,). For example, **195** is in standard form. It can be expanded as **195 = 1 × 100 + 9 × 10 + 5 × 1**.

Astronomers, biologists, engineers, physicists and many others come across quantities whose measures require very small or very large numbers. If they write the numbers in standard form, it may not help us to understand or make computations easily. **Scientific notation** is a way to make these numbers easier to work with.

To write in scientific notation, follow the form $S \times 10^a$. where **S** is a number (integer or integer with decimal) between 1 and 10, but not 10 itself, and a is a positive or negative integer. Thus, a number in scientific notation is written as the product of a number (integer or integer with decimal) and a power of 10. We move the decimal place forward or backward until we have a number from 1 to 9. Then, we add a power of ten that tells how many places you moved the decimal forward or backward.

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